

Sardar Patel Institute of Technology
Department of Computer Engineering | Academic Year: 2025–26

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Course	Computer Engineering – IV
Division	A
Batch	A
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Experiment	No. 11 – D3.js Lab: Building an Animated Bar Chart from Scratch

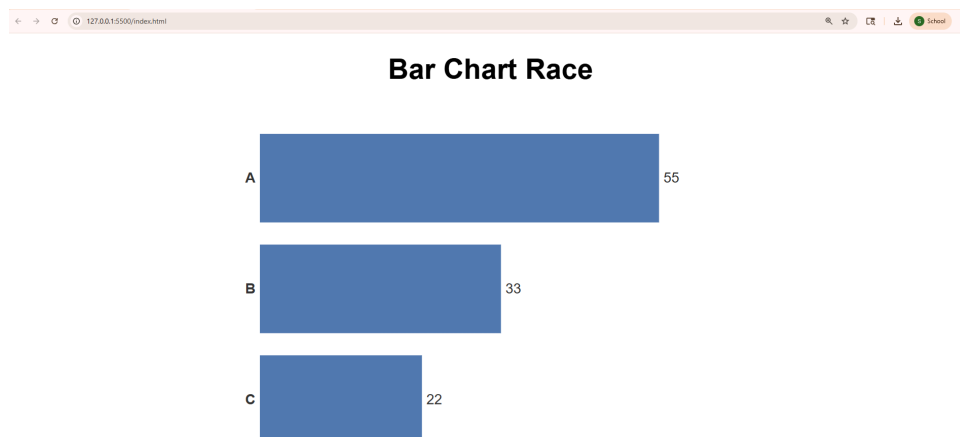
Part A – Animated Bar Chart

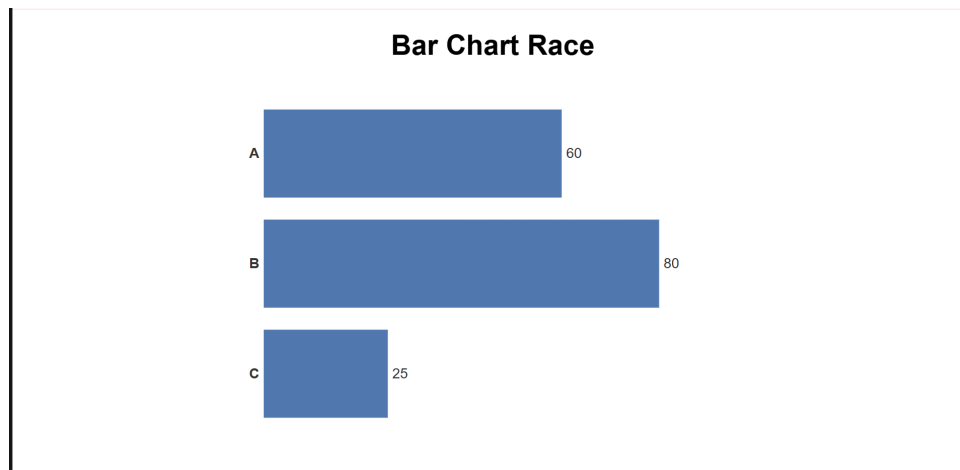
Problem Statement

Build an animated bar chart from scratch using D3.js. The chart should display data for three categories (A, B, C) across years 2020, 2021, and 2022, and animate automatically between years every 1.5 seconds.

Output – Basic Bar Chart (Default Dataset, Steelblue)

The initial output with the default dataset values. Year 2020 shows A=55, B=33, C=22. Year 2021 shows B changed to 80.





Part B – Practice Tasks Performed

Task 1 – Changed Bar Color to Pink

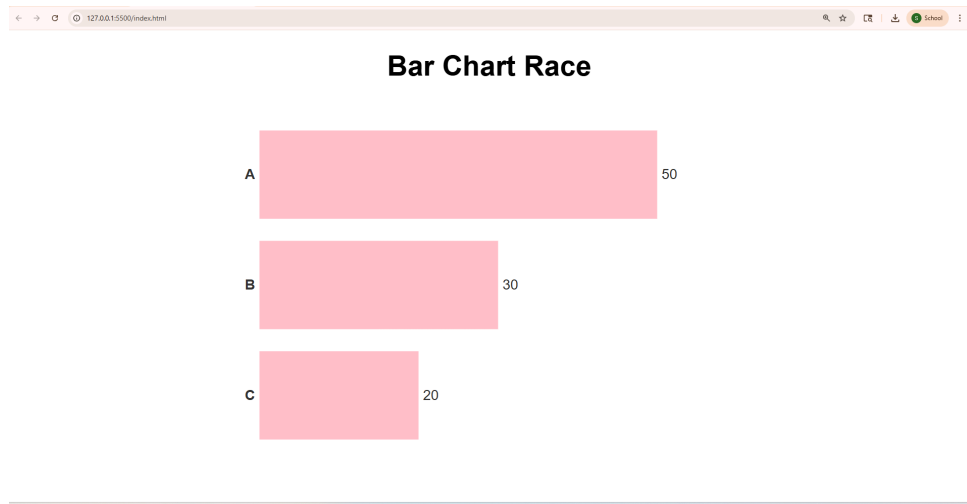
Modified the fill attribute in script.js from "steelblue" to "pink" to change the bar color. The chart now renders all bars in pink across all years.

Code Change:

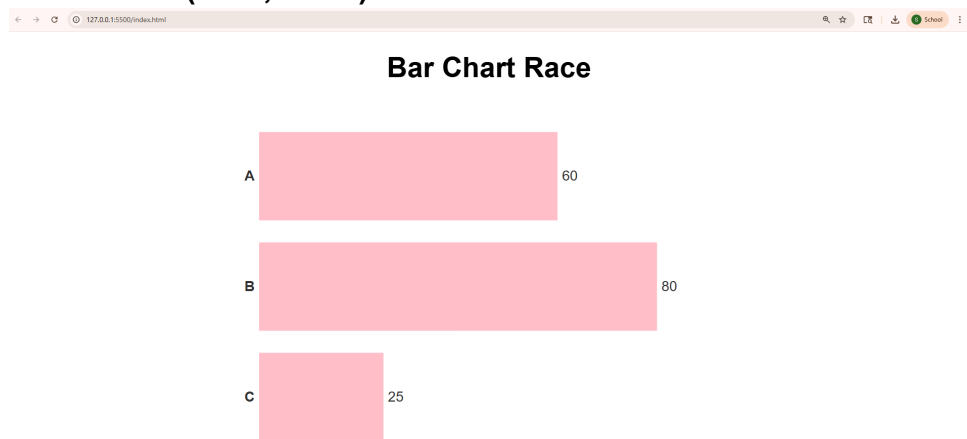
```
bars.enter()
  .append("rect")
  .attr("y", d => yScale(d.name))
  .attr("height", yScale.bandwidth())
  .attr("x", margin.left)
  .attr("width", 0)
  .attr("fill", "pink") // changed colour of bars from blue to pink
  .merge(bars)
  .transition()
  .duration(1000)
  .attr("y", d => yScale(d.name))
  .attr("width", d => xScale(d.value));

bars.exit().remove();
```

Output – Year 2020 (Pink):



Output – Year 2021 (Pink, B=80):



Task 2 – Changed Animation Speed to 3000ms

Changed the setInterval delay from 1500ms to 3000ms so the chart transitions more slowly between years, making the animation easier to observe.

Code Change:

```
const years = [2020, 2021, 2022];
let i = 0;
setInterval(() => {
  update(years[i]);
  i = (i + 1) % years.length;
}, 3000); // changed from 1500ms to 3000ms
```

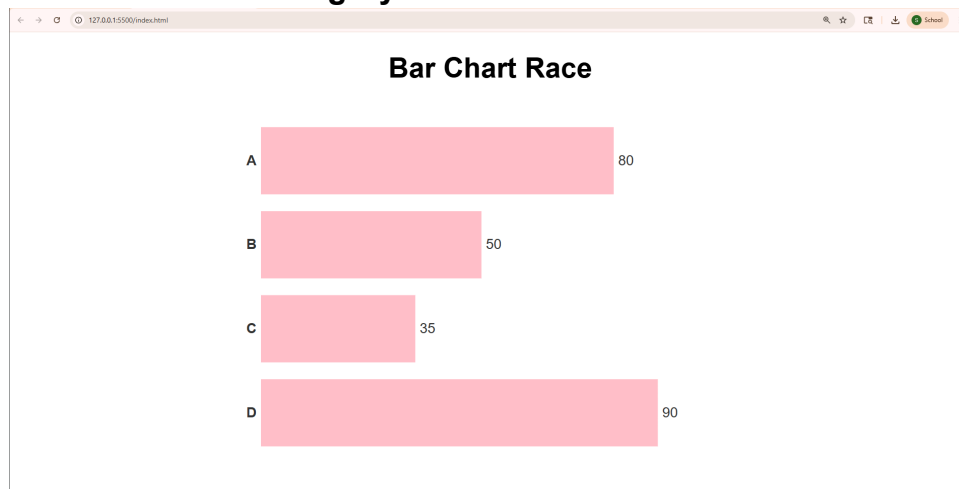
Task 3 – Added Category D to Dataset

Added a new category D to the dataset for all three years (2020: D=10, 2021: D=40, 2022: D=90). The chart now shows four bars animating across years.

Code Change – Updated Dataset:

```
JS script.js > ...
1  const data = [
2    {year: 2020, name: "A", value: 50},
3    {year: 2020, name: "B", value: 30},
4    {year: 2020, name: "C", value: 20},
5    {year: 2020, name: "D", value: 10},
6    {year: 2021, name: "A", value: 60},
7    {year: 2021, name: "B", value: 80},
8    {year: 2021, name: "C", value: 25},
9    {year: 2021, name: "D", value: 40},
10   {year: 2022, name: "A", value: 80},
11   {year: 2022, name: "B", value: 50},
12   {year: 2022, name: "C", value: 35},
13   {year: 2022, name: "D", value: 90},
14 ]; // added category D
15
```

Output – Year 2022 with Category D:



Part C – Swiggy Data Dashboard

Problem Statement

Load a real-world CSV file containing Swiggy order data using PapaParse and build an interactive dashboard using D3.js. The dashboard should display multiple charts arranged in a grid layout.

Charts Built

1. Top 10 Cities by Orders – horizontal bar chart showing Bengaluru leading with ~20,000 orders.
2. Orders by Month – line chart from January to August.
3. Rating Distribution – vertical bar chart showing most orders rated between 4 and 4.5.
4. Top 10 Restaurants – horizontal bar chart with McDonald's at the top.

Output

